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Design fiction as critical pedagogy for exploring technology futures in social research

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This article argues for and demonstrates the application of design fiction in educational settings to engage an interdisciplinary group of participants from civil society and from academia in an educational setting in critical reflection on digital technology futures. It draws on insights from a series of design fiction workshops that we have facilitated with students and researchers, activists, artists, and other members of civil society in Denmark, Norway, and France. In these workshops, participants imagine, prototype, and critique imagined futures where technologies such as electronic monitoring and facial recognition serve as innovations in domains such as employment, sexual health, and tourism. We demonstrate how humor and playfulness function performatively to enable participants to embody design fiction scenarios and convey these dramatically. The workshops offer key pedagogical opportunities to raise ethical questions about technology futures, encourage self-reflexivity, and reveal to participants the limits of their imagination.

KEYWORDS
design fiction, digital detention, ethics, futures, surveillance, technology

1 Introduction

Artificial Intelligence, surveillance tools, and automated and robotic technologies are fast permeating everyday life for people around the world (Jandric and Hayes, 2022). The pace of digital technology development is fast outstripping the pace in which people affected can understand and shape its future. There is also an overall concern with the future on the rise. Catastrophes such as climate change, wars, a global pandemic, political polarization underline the pressing need to imagine alternative futures. Developing critical awareness and curiosity can be useful in pushing back against the seeming inevitability (Zenz and Powles, 2024) of emerging technologies, driven by visions of calculable futures (Williamson and Komljenovic, 2023). There is a need to cultivate our capacity to imagine alternative futures to tip the scale toward the future outcomes that we prefer. We suggest that design fiction provides a useful avenue for doing exactly this.

This article argues for and demonstrates the application of design fiction in educational settings to engage an interdisciplinary group of participants from civil society and from academia in critical reflection on digital technology futures. Design fiction is an approach that blends elements of design, science fact, and science fiction (Bleecker et al., 2022),

to stimulate thought, foster discussion, and encourage reflection on the implications of potential futures of emerging technologies, products, or services on society (Williamson et al., 2023; Dunne and Raby, 2024). As a critical practice it aims to raise questions about how technologies could and should come into being and in the future, highlighting potential problems and opportunities. Thus, its main goal is encouraging critical reflection and discourse rather than designing realistic prototypes or finished products—design for critical debate rather than design for actual production (Knutz and Markussen, 2023).

Since 2023 the authors of this article have been organizing a series of design fiction workshops that with students and researchers, activists, artists, and other members of civil society in Denmark, Norway, and France. During these workshops, we have been using a design fiction toolkit inspired by one of the authors, Sanchez Boe's anthropological research on the use of detention technologies in the policing of migrants and asylum seekers in the USA (Boe and Mainsah, 2021). Using various adaptations of this toolkit, we have been prompting our workshop participants to imagine, prototype, and critique futures where such detention technologies appear as solutions to societal challenges such as population control, health, labor, and tourism. In these workshops, our goal has been to introduce participants, mostly comprised of social scientists and non-designers, to the experience of making design fictions as a mode of speculative social research.

Drawing on samples of fiction, the authors observations (in their position as facilitators and researchers), and post-workshop debriefing discussions with participants, the article explores what social researchers and students might learn from participating in the making of design fiction, and its potential as a form of speculative social research. The article contributes to the growing volume of research on design fiction, and speculative inquiry in general, in media education (Wuyckens, 2021, 2024), engineering education (Hervé and Huez, 2025), design education (Ward, 2013; He et al., 2025; Morrison, 2016; Encinas et al., 2023) sociology (Lupton, 2018; Halford and Southerton, 2023) and science and technology studies (STS) (Meskus and Tikka, 2024). It responds to interest within social science circles in the development of future-focused qualitative methods (Pink, 2025; Watson, 2025)

2 Background

2.1 Design fiction

Design fiction is an approach that brings together elements of design, science fiction, science fact, along with futures studies to stimulate critique and encourage reflection on the implications of potential futures of emerging technologies, products, or services on society (Li, 2024). The term "design fiction" first appeared in Williamson et al. (2023) book "Shaping things" and was later taken up and refined by Bleeker (2009) and Kirby (2010). Design fiction is essentially a speculative and narrative design approach that combines prototyping and storytelling to envision future worlds. Bleeker (2009, p. 6) defines design fiction as "a conflation of design, science fact, and science fiction", and Sterling defines it as "the deliberate use of diegetic prototypes to suspend disbelief about change" (Bosch, 2012, n.p.). Diegetic prototypes are design elements or artifacts seamlessly integrated into the fictional world. They foster a sense of realism and help immerse the audience in a believable narrative experience (Li, 2024). Diegetic prototypes are created by combining two activities: prototyping and storytelling (Wuyckens, 2021). Thus, as Wuyckens suggests, design fiction is about telling stories via and with created objects, seen as artifacts of a story world. The stories provide context and meaning by bringing people and social lives to light, whilst the objects represent embodiments and applications of the envisioned technology, product, or service that transport the audience to imagined future worlds. Lindley and Coulton (2015) propose a definition of design fiction that comprises of three components: "(1) something that creates a story world, (2) has something being prototyped within that story world, and (3) does so in order to create a discursive space" (pp. 210–211).

We are particularly interested in one type of design fiction—participatory design fiction, which combines design fiction with the principles of participatory design (Darby, 2023; Ambe, 2020; Nägele et al., 2018; Oliver, 2019; Duggan et al., 2017). Sanders makes a persuasive argument for participatory design fiction in the following terms:

> We can create and use methods of participatory design fiction to reach toward collective dreaming. I am not talking about situations where designers and futurists create future scenarios/environments for others to experience and think about. I am talking about extreme collective dreaming where all the people can collaboratively imagine, create, and then enact their own future scenarios (Sanders, 2017: 213)

As participatory practice, design fiction can contribute toward efforts at democratizing earthly futures (Urry, 2016). In most societies, the task of future-making often lies in the hands of specialists and experts, leaving out ordinary citizens. The task of democratizing futures therefore responds to the challenge of involving planetary publics in deliberations about future-directed planning, policy, and discourse. As a way of engaging publics with science and technology, design fiction can serve as a way of identifying their understandings of modern technologies and the benefits and risks they see associated with these technologies in future worlds. Design fiction follows the approach of other speculative design practices that utilize literary devices such as satire, exaggeration, and allegory (Malpass, 2013) to provoke imaginative responses. Design fiction invites participants to craft or respond to speculative artifacts and narratives, provoking them to question conventions and assumptions, and to think differently and creatively (Lupton, 2018). Different kinds of tools have been used to introduce participants to design fiction, commonly incorporated in collaborative workshops using toolkits such as "The Thing from the Future" (Candy and Watson, 2015), "The Futures Bazaar" (Candy, 2022), "2023 Work Kit of Design Fiction" (Bleecker, 2024), "Future Philosophical Pills" (Fuel4Design, n.d.a), "Nature Cards" (Rodgers et al., 2023) and "Virtual Design Workspace" (McVeigh-Schultz et al., 2018).

2.2 Design fiction as pedagogical practice in social research

Futures have become a major focus of social research, as scholars interrogate how to relate to increasing threats to social life, caused by rapid technological change, rising inequality, war, political instability and the climate crisis (Urry, 2016; Benjamin, 2024; van Emmerik et al., 2025; Halford and Southerton, 2023). Halford and Southerton (2023) have called upon sociologists to constructively challenge future-making routines, disrupt dominant narratives, and collaborate in building new ways of thinking about and doing futures (Halford and Southerton, 2023: 27). Similarly, Watson has echoed calls by other social researchers to conceive "new visions of the future, for counter-visions, reimaginings, for stories that speak back to power and critically interrogate how futures are seen" (2025: 56). However, the methodological conditions under which such speculative endeavors might happen have not been fully explored. In the face of the types of challenges to the social world enumerated in the beginning of this section, some scholars claim that it is a matter of urgency that student-centered pedagogies include "futures literacy" (Häggström and Schmidt, 2021). Futures literacies imply developing:

> the capacity to imagine a future that is radically different from what we know to be the prevailing order of things, is a way of breaking through the barriers of convention into a sphere of the imagination where many things beyond our everyday experience becomes possible (Friedmann, 2002: 103)

Futures literacies refers to "the ways in which we perceive, sense, enact, envision, and create the future in the present" (Horst and Gladwin, 2024: 42). Horst and Gladwin take up futures in the plural form to highlight the fact that futures contain many meanings, and to indicate that futures are open and undetermined. Futures literacies imply a plurality of interdisciplinary perspectives in future-oriented practices, and foci of futures thinking, brought into a complex assemblage of pathways to understand, perform and diversify futures yet to be determined. The future is a space of projection, which can be filled with ideas and promises, used to mobilize support for or against certain visions. Potential futures can be both enabled and constrained materially and imaginatively by existing structures and discourses that enable people in positions of power to shape narratives and visions of the future—"to discursively and materially "close down" or "open up" futures that serve their vested interests" (Barendregt et al., 2024: 125).

It is within this context that design fiction might offer an avenue for social researchers to explore the ways in which futures are envisaged and enacted. Design fiction enables useful forms of narrative learning for those who participate in it. Narrative learning builds on the idea that humans naturally perceive their experience through narratives, which serve as cognitive frameworks, and tools for framing, interpreting and communicating their perceptions of the world. Narratives enable people to understand and make sense by contextualizing abstract ideas and concepts and providing a scaffold for knowledge in situated contexts (Tanenbaum, 2014). Design fiction is, at its core, about telling stories with and through designed products in imagined future worlds. It allows participants to construct rich story worlds complete with characterizations and intricate plots, providing a rich space to convey and explore intended and unintended consequences of future technologies (Li, 2024). The stories provide context and meaning by situating people and their everyday lives within them, while exploring the imagined consequences of technologies, products, or services. Design fiction shares the design discipline's strength for making things tangible. This implies that design fiction can contribute to making abstract imaginaries of the future of technologies more concrete and accessible, enabling us to contemplate consequences and implications (see Ward, 2021).

Design fiction might contribute to the development of technological fluency for students and scholars in non-technical disciplines by offering a better understanding of its uses and effects, and by providing "providing them with a new perspective from which to consider the discourse about technology they encounter in everyday life" (Lukens and DiSalvo, 2012: 24). In the practice of design fiction, we identify elements of critical making (Ratto, 2011), critical thinking, and critical imagining (Malinverni et al., 2025). Critical making is a practice that merges critical thinking with hands-on experiments to encourage learning by doing. Drawing on constructionist approaches to project-based learning critical making aims "to use material forms of engagement with technologies to supplement and extend critical reflection and, in doing so, to reconnect our lived experiences with technologies to social and conceptual critique" (Ratto, 2011: n.p.). Regarding emerging technologies, critical thinking involves fostering learners' capacity to critically analyze technologies that surround them as well as visions of technology futures. It involves questioning pre-established assumptions, premises, and discourse embedded in these technologies. Critical imagining involves the ability to challenge existing paradigms and envision unestablished and unconventional possibilities. It involves imagining other possibilities and other possible worlds. It implies breaking up with the established ways of understanding, designing, and using technologies (Malinverni et al., 2025).

Light (2021) observes that in much of speculative design work, artifacts made to inspire reflection about alternative futures usually reflect the designer's specific concerns. This narrows the space of alternative visions or interpretations by others. This helps to "shed light on design as an essentially political activity—how design functions as a mechanism for choosing and materializing futures, an act that is neither neutral nor innocent" (Bendor and Lupetti, 2025: 406). Students may develop critical reflexivity on specific design practices and roles, the values that orient design, and contemplate the designer's agency and responsibility (Encinas et al., 2023). As Encinas et al. (2023), suggest:

> The processes of creating speculative designs can encourage interrogation of prevailing assumptions and invite exploration of other, alternative states of being and doing. These activities can lead to a deeper understanding of, for example, the contextual, political and cultural factors that influence the activity of design, and in turn, consideration of the potential implications and effects caused by bringing new products and services into the world.

Thus, students engaging with design are likely to develop a more sophisticated understanding of the designers' role as "agents of the public imagination" (Farias et al., 2022: 147). From the position of maker of future worlds, students can be encouraged to think differently, considering values, and practice in general in more sophisticated ways. Weaving together speculative technology artifacts and stories around them can enable students to engage with ethics in a visceral, embodies, practical way, drawing together "head", "hand" and "heart" (Bendor and Lupetti, 2025: 407), in a process grounded in materiality, interactivity and relationality.

3 Materials and methods

3.1 The toolkit and the workshops

To explore the role of design fiction as a pedagogy to explore speculative approaches to technology futures we have conducted a series of workshops in Aarhus, Oslo, and Paris. In these workshops we have introduced participants to design fiction and engaged them in hands-on group activities to produce design fiction, using a card-based toolkit. The design of these workshops builds on problem-based-learning (Nordin et al., 2024), experiential learning (Blunsdon et al., 2003), cooperative learning (Johnson et al., 1991), and design studio pedagogies.

The toolkit originated as a product of article co-author Sanchez Boe's research on digital detention and immigration control practices through the use of digital confinement devices such as electronic tagging or facial recognition apps (Boe, 2020, 2023; Boe and Mainsah, 2021). The toolkit was designed by co-authors Lippera and Kerspern, in collaboration with Mainsah and Sanchez Boe. It aims to serve as a device for developing speculative scenarios where we imagine digital detention technologies being used in the future to address societal issues in other domains.

The toolkit aimed at aiding workshop participants to develop narrative scenarios and prototypes, essential components of design fictions. The toolkit consists of a card deck, a scenario canvas, and templates of technology device interfaces.

The card deck is comprised of three categories of cards, (a) technology, (b) deployment purpose, and (c) opposition strategy. Each card contained a keyword and a short description.

The seven "*technology*" cards included electronic tag, geo fencing, facial recognition, iris scan, microchip implant, crossbreeding of existing techs, and new/future tech.

The *deployment purpose* cards are based on the seventeen UN Sustainable Development Goals (SDG; United Nations A/RES/70/1, 2015) and contain the following aims: (1) No poverty, (2) Zero hunger, (3) Good health and well-being, (4) Quality education, (5) Gender equality, (6) Clean water and sanitation, (7) Affordable and clean energy, (8) Decent work and economic growth, (9) Industry, innovation and infrastructure, (10) Reduced inequalities, (11) Sustainable cities and communities, (12) Responsible consumption and production, (13) Climate action, (14) Life below water, (15) Life on land, (16) Peace, justice, and strong institutions, and (17) Partnerships for the goals (Figure 1).

The *opposition strategy* category included the following cards: escape/avoid, vanish/disappear, protest, fool/cheat, infiltrate, obfuscate, discredit/defame, expose/show/inform, hack/hijack, slow down, disconnect/divide, break/sabotage, link/collaborate, reverse monitoring, other, each describing a possible strategy to resist the deployment of a particular technology (Figure 2).

The scenario canvas consists of an A3 size printed sheet of paper with boxes where participants place their chosen cards on, describe the key features of their scenario, and list which societal actors would benefit and suffer from it.

The first round of the workshop called "for the greater good" is carried out in the following manner:

1. The facilitators introduce the participants to the genre of design fiction illustrated with examples, describe the workshop activities, and demonstrate how the toolkit works.
2. Working in groups, participants proceed to the first round of speculative work. They are asked to get into the role of "pro-tech partisans" and pick one card from the *technology* category and one card from the *deployment purpose* category and combine the two to generate an idea for a design fiction scenario in an imagined future. Using the canvas sheet the group sketches out in words and/or by drawing the key features of the solution. On the left side of the canvas, they write down who you think would benefit from the solution, why, and how. On the right side of the canvas, they write down who they think would suffer if this solution was implemented, why, and how (Figure 3).
3. Based on the scenario written out on the canvas each group is invited to prototype a communication support or strategy, using the templates and other visual aids at their disposal. The prototype is meant to represent a tangible manifestation of an artifact from the imagined future scenario they have written down. This communication artifact might take off an imaginary press release, the interface of a mobile device, a customer review from a user, or a street advertisement board.
4. Each group presents their design fiction to the rest of the workshop participants. This opens for a general discussion on the ethical, social, and political implications of fiction scenarios bringing to light issues and controversies.

In the second round, called "taking the power back", participants repeat steps 2–4. In this second-round participants personify the role of "anti-tech opponents," this time combining the card they initially chose from the *technology* category and one card from the *opposition strategy* bunch to develop an idea for a system to resist the future technology created in the first round. Through these two rounds, the aim is to aid participants to develop a "polyphony of visions", to display futures characterized by plurality—of a variety of actors, interests, values, opinions, initiatives. It is to help the audience grasp complexities and collectively think through potential controversies between different stances and points of view.

3.2 Research material and analysis

We have organized six different workshops in which we have used the toolkit described above, in Aarhus, Oslo, and in Paris,



FIGURE 1
Examples of deployment purpose cards.

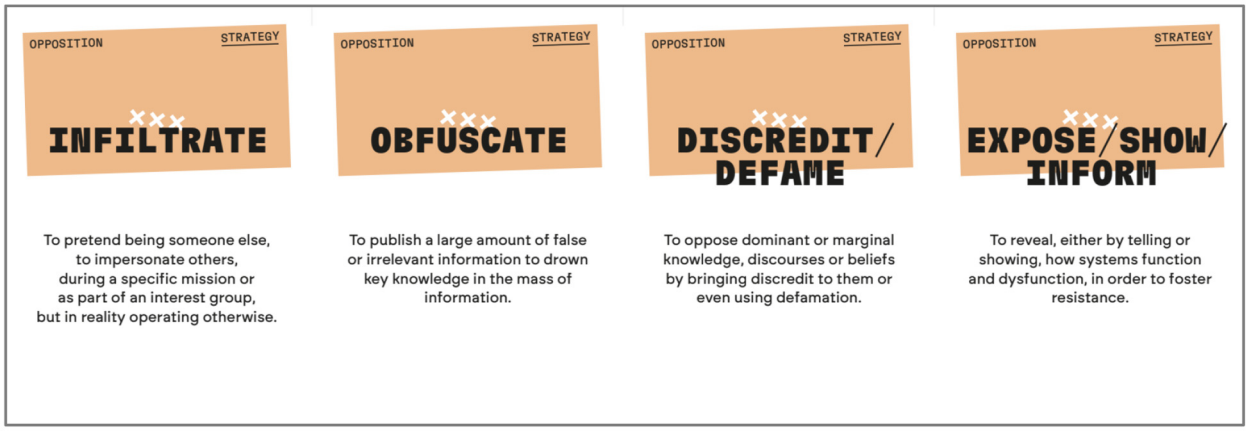


FIGURE 2
Example of opposition strategy cards.

over a period of 18 months, which have generated over 20 fictions. The workshops have varied in length from long formats that lasted 12 h to shorter formats that lasted 4 h. Through these different workshop variations, we have been able to experiment with different constraints in terms of time and to observe the dynamics that emerged when we put together varied sizes of groups.

Our reflections in this article are based on our involvement as facilitators and participant observers in all these workshops. As sources of research data, we have relied on observation notes, discussions with participants, and what the fictions created by participants revealed. During the workshops we moved from one group to the other taking notes and pictures of what we observed, as the process unfolded. We held debriefing sessions with participants as part of the wrapping-up part of these workshops. To prompt reflection during these debriefing sessions with participants we asked them how they experienced the activities, what they learned, and how this could be applied in their future research and studies. After each workshop, we also debriefed with one another as part of the facilitator-researcher team, reviewing the outputs of the workshops and our observation notes. We discussed challenges faced, explored our understanding of the participants' experience, and developed suggestions to improve the process.

4 Results

In this section we will show four examples of the design fictions produced by participants from two of the workshops. The fictions that we present below are not meant as representative of a larger sample, but are rather meant to serve just as examples of what the fictions produced in the workshops looked like with a high level of granularity. What we were interested in were not the details of the narratives, *per se*, but rather what they indicated about the imagination of the participants.

4.1 Aarhus workshop 1

This workshop took place in Aarhus University on 20th–21st June 2023. The purpose of the workshop to reflect together

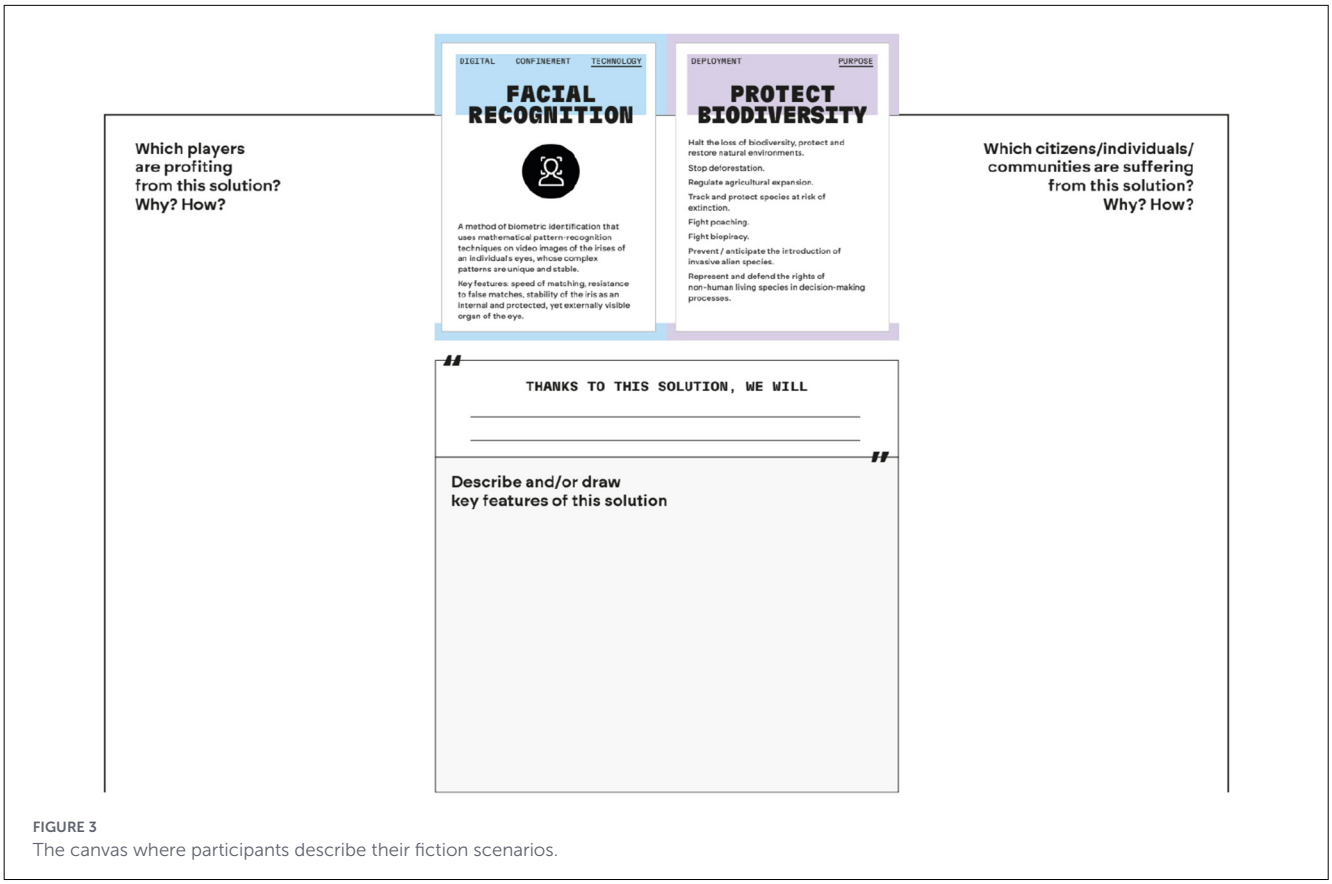


FIGURE 3
The canvas where participants describe their fiction scenarios.

on speculative futures opened by the massification of so-called "alternative to detention technologies"; such as electronic shackles and facial recognition apps which were originally developed by the for-profit prison industry, and increasingly deployed in immigration enforcement contexts, extracting value and data from both criminalized citizens and illegalized migrants (Boe, 2020, 2023).

What if, tomorrow, these digital confinement technologies were to be massified and normalized until being deployed against anyone in their everyday life? As a starting point we gave the participants the following prompt:

> *In 2033, what if containment and biometric technologies had spread from the prison and immigration enforcement environment into other societal contexts leading to the surveillance, control, and exploitation of (more and more) people?*

The group was selected to include 12 participants with a variety of backgrounds in terms of age, gender, interest for, use of, and knowledge of technology, and experiences of living for longer periods of time in different countries (Denmark, Switzerland, Germany, France, Norway, Cameroon, Nigeria, the UK, Cuba, the USA, among others). Social science researchers included junior and senior scholars, master- and doctoral-students, lecturers and professors.

4.1.1 "SOME MEN CARE"—EVE

Cards chosen:

Technology: Microchip implant (Digital Detention Technology
Deployment purpose: Achieve gender equality, Protect health)
Aim: "Thanks to this solution, we will provide menstrual justice for all."

Fiction: *EVE is a solution developed by men for women, to help women monitor and regulate their hormone balance to avoid suffering from menstrual pain and avoid hysteria (announced with much irony by the group made up of two men and a woman). EVE, developed by the company "Some Men Care" is a solution which relies on a microchip implanted in women's bodies, delivering electronic impulses, without any input of chemicals or external hormones. EVE makes it possible to avoid dealing with things such as menstrual leave that are costly for companies and for the State. It reduces the risk of "menstrual fraud" by women who pretend to be suffering from menstrual pain and call in sick. (EVE's marketing rep insists on this point with a healthy dose of sarcasm). EVE guarantees women who use it a "stable" and "good mood", with positive effects for the family, and society at large. Mothers can implant EVE chips into their daughters' bodies from the age of 14 years old, or even earlier, depending on the first menstruation cycle. The day this happens becomes a special moment in a girl's life, a ritual bonding daughter and mother together. What if EVE becomes a prized gift that lovers can offer each other, as advertised on street billboards with titles*

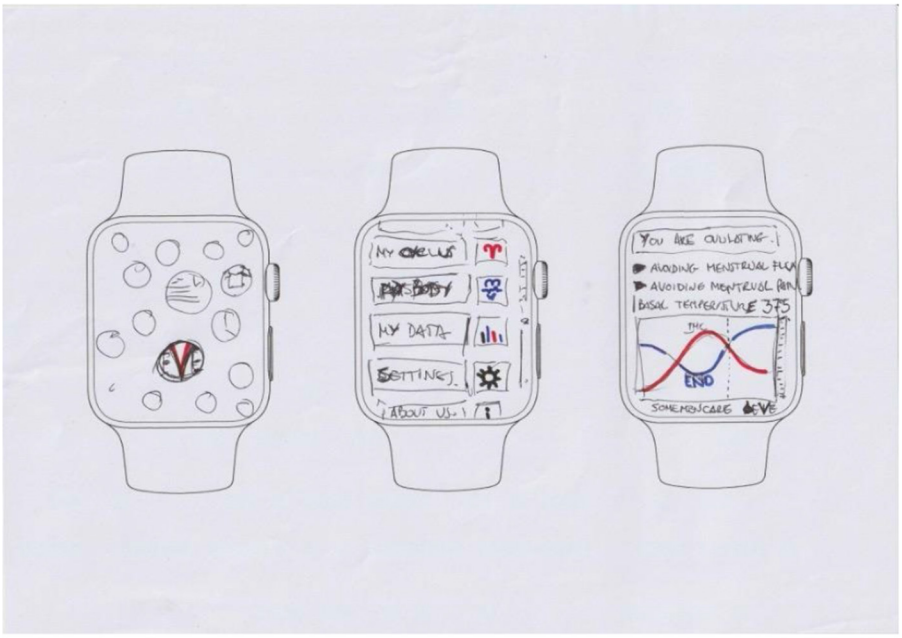


FIGURE 4
EVE app open on a smartwatch showing menu functions and visual display.

such as "Some Men Actually Care—For Valentine's Day, Offer Her Control, Offer her EVE" (Figure 4).

After EVE goes out into the world, not everyone supports this innovative technology. EVE deals a severe blow to the female hygiene product industry, and women no longer need sanitary products. The company proceeds to work on future applications of their invention in the fields of mental health, to treat addictions, overweight, reduced productivity, sex, and libido, as well as mood management.

Opposition strategy: *In the aftermath of EVEs appearance in the market protest movements spring to action proclaiming slogans such as "My body is sacred," "Give me my period back", "Leave my periods alone", and "Natural pain matters". Protesters use a variety of opposition strategies such as showing users how to supply false data into the chip inserted in users' bodies, remove implanted chips, hack the system, and even reverse surveillance by using the chip to monitor big tech companies in return. Data and privacy rights advocates try to warn about risks associated with this digital solution. They warn that there may be a risk that anti-abortion movements could hack the data and extract info about ovulation cycles to identify women having had (illegal) abortions. Some women cannot afford EVE and rely on solidarity groups to buy the chip* (Figures 5, 6).

4.1.2. "JOB4SURE"

Cards chosen:

Technology: Electronic tagging and Geofencing

Deployment purpose: Reach full employment.

Aim: "JOB4SURE will help job seekers identify and actively search and apply for jobs and enable. employers find relevant employees easier".

Fiction: *An app called JOB4SURE appears in the world. It is the result of a collaboration between governmental agencies and tech companies. It represents a revolution in the job search market. The app enables job seekers to receive all-day notifications of job alerts in their area and prompts them to move to the location of the job to apply. Users get points for each job they apply to through the app. When they reach a certain number of points, they receive a reward: they get free public transport—aimed to ease their job search and make them apply more.*

JOB4SURE helps match job seekers with opportunities waiting around and encourages movement to make sure job seekers remain active. The app can be accessed through discrete and casual wearable devices, such as smartwatches, so its unemployed users are not stigmatized.

Employers get better access to the workforce—it comes to them—and are encouraged to offer new types of jobs and micro-jobs. For some communities, the solution is viewed positively as a tool to bring more social contact and cohesion during job seeking periods, more gender equality—working from home, with kids, cooking in the kitchen would be considered proper jobs—and enhance environmental justice by encouraging daily missions such as cleaning the streets (Figure 7).

On the other hand, the many downsides of this solution are quickly spotted and experienced by job seekers: loss of freedom and autonomy, physical and mental stress due to constant alerts, tracking, suspicion of profiling, discrimination, uncertainty regarding what happens with the collected data.

Opposition strategy fiction: *in the aftermath of JOB4SURE being introduced, there soon emerges a resistance movement that calls itself NOT-SO-SURE, gathering employers, job seekers and interest groups willing to take part in a more humane and fair job market.*

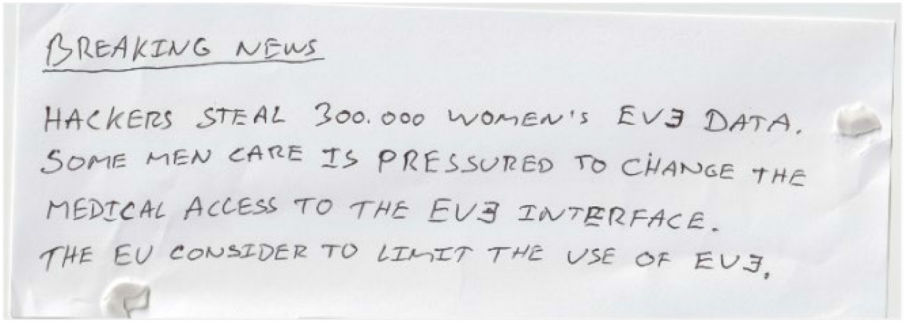


FIGURE 5
Breaking news post: hackers stole data from 300,000 women using EVE.

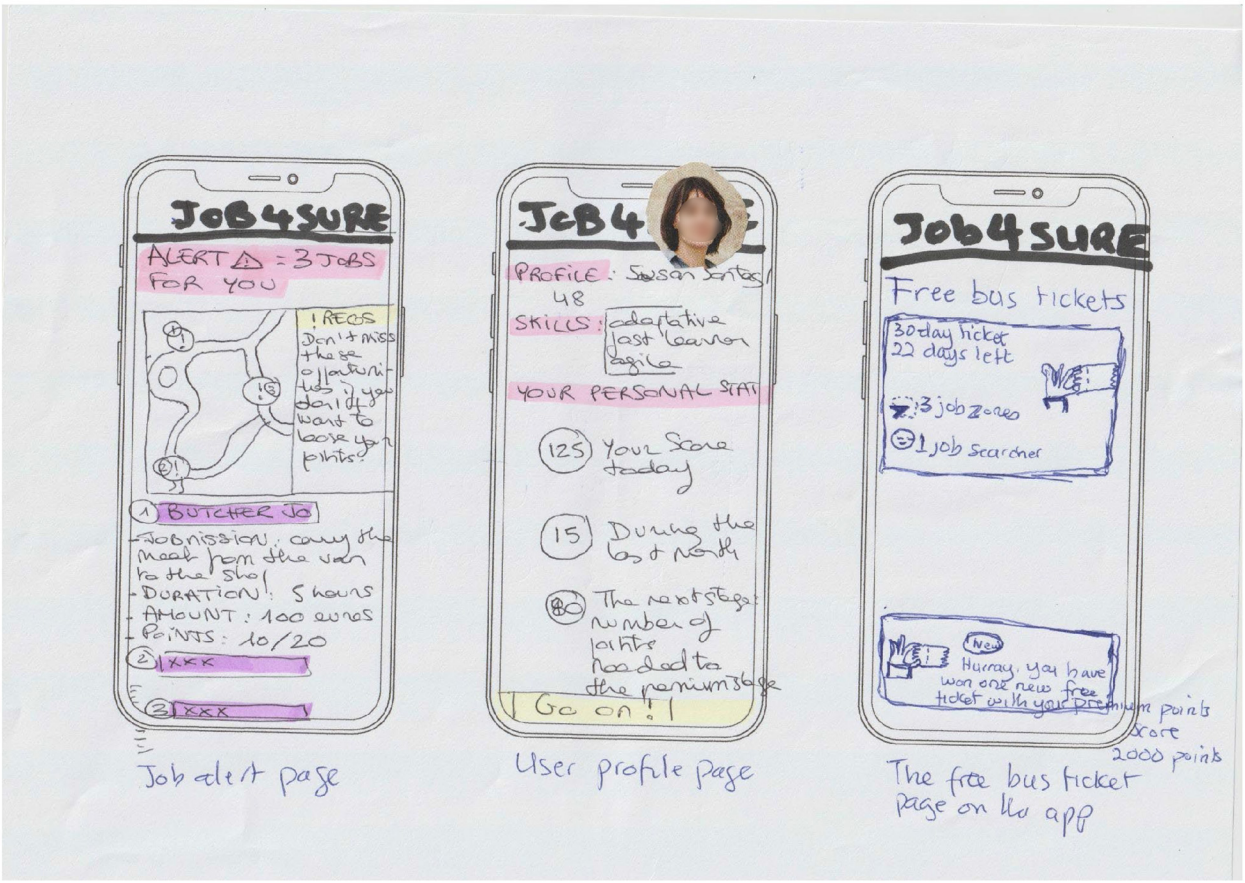


FIGURE 6
JOB4SURE interfaces on a mobile device: (l–r) the job alerts page with a map showing where job opportunities pop up, the user profile page, a page showing a free bus ticket won as a reward.

They start as a discrete and undercover initiative relying on specially designed assets: a coded message logo, a non-permanent tattoo, stickers to stick on the walls of cafes and bars, hidden messages in gardens and parks to transmit information (as the old Telegram app isn't safe anymore in this future). Over time when it reaches a critical mass, the movement goes public, documenting and exposing the abuses of the system of tracking and coercing job seekers through the digital device. They create a network/community of supporters/activists/foot soldiers who organize public events to raise public support and sensitize citizens about the dangers of the job search system.

4.2 Paris workshop 2—The right to the city

The aim of the workshop was to engage students at the Paris Cité University located in the Saint-Germain-des-Près neighborhood to speculatively imagine the deployment

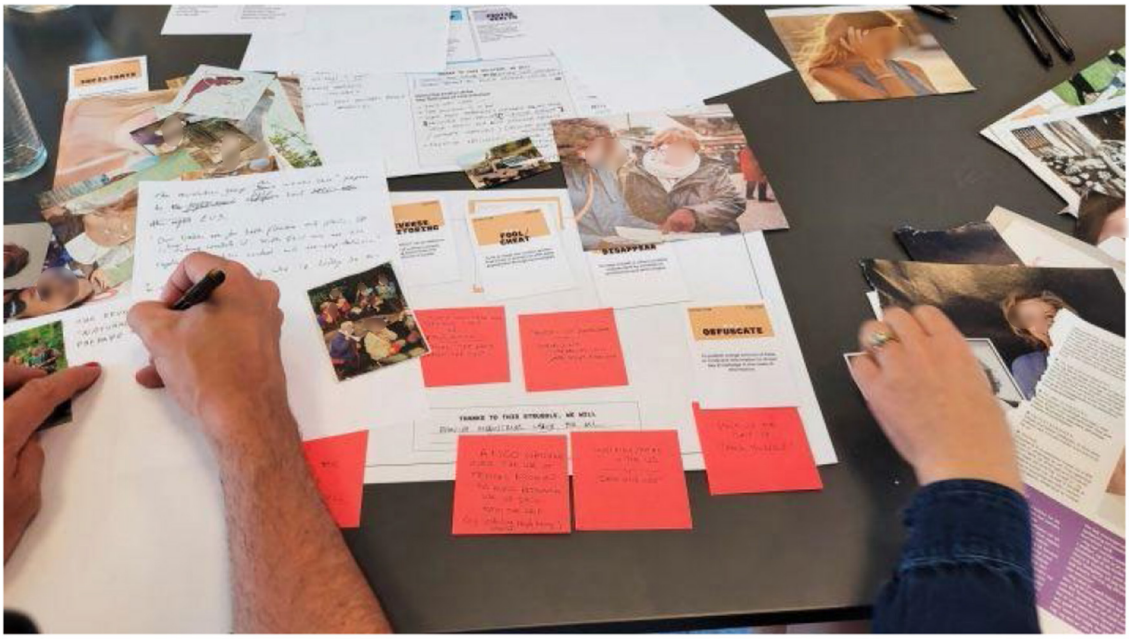


FIGURE 7
A table in a workshop with participants using workshop materials to develop fiction scenarios.

of technologies to address issues related to the right to the city (Lefevre), and particularly, access to the neighborhood, in which they study daily but none of them live. All students, regardless of their research interests, are affected by overtourism in the Saint-Germain-des-Près neighborhood. Besides overcrowding, the commodification of public space and heightened prices for locals, students are sensitized to other problems related to the environment and the overuse of public space related to overtourism.

Our prompt for the workshop was: *what if, in 2030, the municipality of Paris decides to deploy technologies to regulate the flow of tourists in the Saint-Germain-des-Près neighborhood?*

4.2.1 TRS "The Tourist Rating System"

Card chosen *Technology*: Cross Breeding of Technologies

Fiction: *to fight against the problems of overtourism in Paris, the Tourism Rating System TRS is created to police the movement of tourists and to encourage civil behavior. The system is designed to work as follows. Before you arrive in Paris, you will have to watch a video on an online platform set up by the Paris Town Hall, Mairie de Paris, and take a test to prove that you have understood all the rules of good behavior. If the online test shows that you didn't watch the video, you will not be allowed to enter certain neighborhoods. Following the test results, you get points and are rewarded with one to five stars. The test determines your initial mobility, which also depends on how many other tourists are already present in the area. If the area is very crowded, the system will accept only four- and five-star people. If you are a tourist with a low score, you may not be allowed to take more than five photos, but the more stars you have, the more photos you are allowed to take. You can choose to use points earned during the online test, but your initial score does not determine your overall tourist experience, as you can always increase your score with small actions, like helping out a resident or picking up trash in the city. The points determine how many hours you are allowed to stay in certain neighborhoods. The system is able to connect to all the phones, including yours, in a specific area, extracting data from your and other tourists' phones and GPS, mobilizing public facial recognition surveillance cameras, which are then connected to a system that analyzes tourists' individual behavior, supervised by Paris municipality staff.*

Tourist Anna has succeeded the online test and behaved flawlessly: she has the maximum five stars and may stay in the area for an hour, while Tourist Tom smoked in the street and threw his cigarette butt on the ground, so he has only one star left. Hence, he can't stay in the neighborhood, and his presence makes a very noisy alarm go off.

As a tourist, you benefit from the system, because you get a more qualitative experience enjoying a visit with fewer other tourists and better photo opportunities. You will no longer have to wait for a table or queue in front of a shop. Only tourists with good behavior may stay in the neighborhood for longer periods of time, which means less noise and a safer environment for all, a less stressful experience for you, other tourists and locals alike. The university's students, faculty and administrative staff also benefit from TRS because it gets easier to walk between the metro and the university, and around the university. There is an overall improvement of the environment, with less pollution, less visual pollution, less trash in the streets.

Opposition strategy fiction: cards selected: Hack/Hijack, "Break/Sabotage":

Some tourists experience more stress during their holiday, because of being constantly controlled by facial recognition cameras

and municipal staff, and influencers or others who make a living from creating visual material are no longer able to take a lot of photographs or videos. There starts to be pushbacks from civil society groups concerned with privacy issues, data collection and freedom of movement. A movement develops that aims to first disrupt then destroy TRS, the Tourist Rating System. A first wave of resistance is launched that hacks into the application and hijacks the system to send messages to Parisian residents and tourists, warning them about freedom and privacy protection. The activists send messages explaining to users how to hijack the system and disrupt the star accreditation system, making it give five stars to everyone. As a result, the Mairie de Paris creates a new application, which is protected against hacking. We then see a second wave of resistance to ensure that the surveillance system cannot work anymore, through sabotage, where cameras are destroyed, leading to public, collective protests, increased social consciousness, a revolution, and freedom.

4.2.2 "The Paris Bracelet"

Cards chosen: *technology*—cross breeding of technologies.

Fiction: *The Paris Bracelet is developed to address the problem of tourists spending their money in multinational food and retail stores and entice them to spend more in local, independent and Parisian owned stores. Tourism should profit other businesses than the LVMH group and luxury brands or multinational retail chains that have invaded the area over the past two decades and are already very rich, and which do not have a positive impact on labor, the local economy or the daily lives of Parisians. The Paris Bracelet connects to a point system, starting all tourists at 50 points. The bracelet is connected to the GPS network and to street surveillance cameras. If you exceed time, the GPS bracelet rings very loudly and you have to leave the area or you lose points, and ultimately, if you don't respect the rules, you have a punishment. We will strap the GPS bracelets on tourists that arrive at the border by plane. If you take the bracelet off, you lose, and the punishment is stronger if you try to take the bracelet off. As a tourist you will earn and lose points according to the places you visit and whether you spend time and money that benefit the local inhabitants. You get access online or through an app to a list established by the Mairie de Paris (Paris town hall) of places to visit, with information that guides you through the city. Just by visiting marginalized unpopular businesses you gain points. With the points, you earn the right to go to certain places, gaining access to the Saint-Germain-des-Près area, for instance. If you visit Saint-Germain-des-Près and respect your time and keep your score perfect, and if you leave Paris with a positive score, you could get x% at the duty free or a gift, such as three macaroons from an independent Parisian manufacturer of pastries." The new system slows down tourists and takes stress off tourist sites and businesses. The Café de Flores will appreciate that their clients stay for a few hours instead of buying one coffee, taking a gazillion photos, then leaving the place. Also, the system aims to avoid the buying up of property by exterior investors who rent apartments out to tourists instead of renting them out to Parisians, thus pushing real estate and rent prices up. Tourists are encouraged to rent independent boarding owned by Parisians who are themselves on holiday, or private apartments and houses located outside of Paris, in the Ile de France region, to spread tourists out from Paris into lesser populated areas of the region.*

Opposition strategy fiction: cards selected: fool/cheat, obfuscate, disrupt, hack/hijack

Opposition strategy fiction: *in the aftermath of the implementation of the Paris Bracelet, tourists start using other airports to enter France and then join Paris by land. Tourists organize and start spreading misinformation. They publish a false Instagram post supposedly from Mairie de Paris requesting large groups of tourists go to one place at the same time. This forces the Municipality to punish large groups of tourists simultaneously. This overwhelms the system making it award excessively large numbers of negative scores. Tourists bubble with indignation and start mobilizing for collective action to boycott Paris or France as a destination. Corporate actors are unhappy at the fact that the Paris Bracelet seems to be supporting the local economy, leading to a fall in sales among their multinational retail and food stores.*

5 Discussion

5.1 Playfulness, absurdity, and humor

During the participants' presentations of their fictions, we heard a lot of laughter. The participants seamlessly slipped into different roles during their presentations inserting elements of absurdity, exaggeration and humor for dramatic effect. For example, we noted down this exchange from the group that created The Paris Bracelet:

Participant 1: This is why I suggested to electrocute tourists!

Participant 2: What happens if they cut their electronic monitor off?

Participant 3: Then they have to pay! Or they can be sent to prison! Now you're going to accuse me of being a totalitarian again! (*adding, with a smile to her team member*).

In this exchange they are knowingly and readily enacting the role of unscrupulous and evil designers of technology seemingly unperturbed by the societal consequences of the things they make and put out into the world. A few moments later, they would switch roles to personify activists trying to hack and disrupt the Paris Bracelet. Through gestures and tone of voice they would exaggerate and over dramatize, inviting the audience to be part of the story. They used semiotic resources such as the body and sketched out technology artifacts in the process of "making" themselves in interaction with fiction (see Lundström and Olin-Scheller, 2014). Seemingly utopian visions of the future often propagated by corporations through popular media typically showcase idealized aspects of everyday life enhanced by technology (Tharp and Tharp, 2019). Through employing comedic techniques in design fiction, participants can satirize these positive visions by showing "look what can go wrong" scenarios depicting potentially alarming outcomes and cautionary tales (Dunne and Raby, 2024: 43).

During the workshops, performances such as these would generate bursts of laughter from the audience. Despite the seriousness of the themes, the design fiction workshop provided the context for participants to freely, and for once, act out roles

such as the non-ethical other, or the villain. This freedom seemed to liberate participants' creative energy and critical reflection. As the workshop unfolded, we could observe as participants "collaboratively imagine, create, and then enact their own future scenarios" through what Sanders, quoted earlier, calls "extreme collective dreaming" (2017: 213).

Dunne and Raby (2024: 40) point out that "a skillful use of satire and irony can engage the audience more constructively by appealing to the imagination as well as engaging the intellect." In design practice, rhetorical devices such as irony and satire have often been used to embed critique and provide provocation for critiquing designs' forms of critical practice (Malpass, 2013). Watson (2011) argues that the uses of devices such as satire, sarcasm and irony have a thoroughly serious, though under-utilized potential in social research.

Fine and Martin observe that, a "predominant characteristic… of scientific discourse generally is its serious and sedate mien" (1990, p. 89). Throughout the group work, we sensed an atmosphere of excitement, playfulness, and joy. Participants routinely expressed how fun and engaging the collective group work was and how it differed with other experiences of group work where some participants might have difficulties concentrating or may disengage from the collective flow. Everyone was clearly in the moment, enjoying themselves. The role-playing during presentations brought both joy and laughter to the process of discussing pressing ethical and social justice issues and the pitfalls of technological solutions to social problems. Some of the participants expressed surprise at how far their own imagination could go.

5.2 Embedding ethics and articulating self-reflexivity

When we invite participants to weave together speculative technology artifacts and stories, we are implicitly pushing them to engage with ethics in a visceral, embodied, practical way, drawing together "head", "hand" and "heart" (Bendor and Lupetti, 2025: 407). This is a process grounded in materiality, interactivity and relationality. What would the moral ramifications of an imagined future be when inserting microchips into bodies, scanning faces, tracking movement everywhere we go, and living with robots become the norm? Are technological inventions that enable microchips to be implanted in women's bodies acceptable? What are the unintended consequences of policing public space with surveillance technologies, even when done for legitimate reasons? Who profits and who suffers when we make citizens spy on each other? The fictions presented during workshops were replete with ethical dilemmas and provided fruitful ground for further discussions about the unintended consequences of technology futures regarding issues such as privacy, surveillance, human rights, and abusive commercial practices. When discussing these types of questions, we suggest that participants are in effect collectively engaged in what Johnson (2011) terms anticipatory ethics. This is a form of ethics that is enacted when we raise questions about the potential mid- to long-term social, ethical, political, and environmental implications of technologies.

Another issue that the workshops highlighted for us was the value of design fiction in encouraging participants to be self-reflexive about their hidden biases and assumptions. We noticed, for example, the absence of reflection about the environmental consequences of the technologies imagined in the fictions. After such an observation, we can probe participants to reflect on the sourcing of rare earth minerals, plastic used for the manufacturing of technological devices or resources that spur conflicts and war in different parts of the world such as the Democratic Republic of Congo, or water scarcity in Bolivia. This reflects the point made earlier in reference to Light (2021) observing how artifacts made to inspire reflection about alternative futures usually reflect the designer's specific concerns narrowing down the space of alternative visions or interpretations by others. We see the action of inviting participants to play the role of the designer and to develop fictional stories and things as a way of demonstrating that when design functions as a mechanism for choosing and materializing futures, it is an act that is rarely neutral nor innocent (Bendor and Lupetti, 2025: 406).

5.3 Limits of imagination

One common theme emerged during many of the discussion of the design fictions created by participants: Despite concerted attempts at imagining radically alternative futures, most of the design fiction scenarios that the participants created seemed to fall back on familiar tropes. Many aspects of the solutions portrayed in the narratives were based on things that already exist. As one participant from the Right to the City workshop remarked, "we are inspired from things that we have already seen, from reality. Things that we saw during the COVID 19 crisis, for instance, which makes the idea of surveilling tourists and reducing their mobility not so surreal after all". The participant referred to the many months of overnight curfew during the pandemic where the mobility of each Parisian was also reduced to a perimeter of 1 km from their home, and for a maximum of 1 h. To prove at what moment they had left their home, Parisians had to either download a geolocation app on their smartphone or print out a government issued paper form with their name, address and time of leaving home, which they should be ready to present to police officers who controlled some of the most popular neighborhoods of the city, like Montmartre, which is one of the more green areas of the city, and other areas that Parisians missed going to during the lock-down. At the same time, drones with surveillance cameras controlled the banks of River Seine.

Participants routinely encounter endless streams of discourses through advertising, popular science fiction films, books and other media. These discourses tend to depict deterministic relationships between humans and technology that offer no alternative but a future where surveillance technologies are everywhere, where data capture is omnipresent, and where decisions that affect the lives of citizens are increasingly automated. Markham (2021) describes the difficulty people face in imagining futures in ways that do not reproduce current ideological trends or cede control and power to external, mostly corporate, stakeholders as a form of discursive closure. The ability to imagine differently about the near or distant possible futures requires of the people engaged

to break out of existing discursive frames prior to creating new forward trajectories. The ability to imagine differently is hampered firstly because our everyday frames of thinking have become so naturalized and invisible. Markham suggests that this is unsurprising given that our imagination is "locked into a distortion loop of contemporary grand narratives, repeated every time we follow a news story on Twitter, stream a new series on Netflix, get caught up in the promise of the latest new tech development, or otherwise consume another cultural unit of information, a nuance which is difficult to find. The storylines are easily restricted to particular trajectories with particular explanations and limitations of what is probable or possible" (Markham, 2021: 398).

One participant described the experience of creating a design fiction as "scary, in the sense that it reminds us how these new devices and technologies always seem to develop "into despotism, it's as if all the things that we create are dystopian". She added, "I don't know how to transform these devices into something which could be used more positively, to create a utopia rather than a dystopia". "It is scary, but also thrilling" to do a participative design fiction project", she concluded. Another participant remarked how fascinating it is that "All our Machiavellian minds come together to think of something which easily ends up being a terrible dystopia".

McGreevy and colleagues argue that the embodied and habitual nature of everyday culture makes it difficult for us to extract ourselves from the trajectory of experience and commonly held assumptions (McGreevy et al., 2022: 2). It is only when these hegemonic frames of thinking are made visible and disturbed that we notice the role they play in shaping our sensibilities and visions. Secondly, we are so conditioned to articulate predominant thoughts about social progress that attempts at imagining alternative futures tend to reproduce a narrative arc that constitutes a sense of inevitable continuation of what already is.

5.4 Technological fluency and literacies of making

Which functions would an app that guides unemployed people to job locations in a nearby future include? How would a microchip inserted under the skin feel? These are some of the myriads of questions that participants face when creating design fiction prototypes to accompany their narratives. To develop solutions some participants would during the workshops draw on personal knowledge or experience during the workshops. Others would go online and search. A few would at times turn around and ask workshop facilitators or other more technologically knowledgeable participants. They enter the world of design fiction as sociologists and anthropologists trained to discuss the moral and social questions surrounding technologies but often in abstract terms, far removed from concrete designs. Instead of abstract reasoning about potential implications of technological futures, which often might remain aloof and abstract, the process of crafting diegetic prototypes in a design fiction workshop makes futures we imagine more tangible.

The types of issues participants in design fiction workshops engage with require an inter- or transdisciplinary approach, requiring them to draw on knowledge from diverse fields. In the workshops, participants are invited to adopt roles such as, designer, corporate marketing executive, activist, and software engineer At times, we observed negotiations during group work, which occasionally would lead to either unanimously adopting the idea of one of the participants, a compromise, or discarding an idea altogether. During one of the group presentations one participant expressed wanting to create a mechanism that made it possible to denounce other tourist's' behavior, "people could get to spy on each other and earn additional points", but the idea was dismissed by his group. One other participant regretted: "We could have made our system more repressive, but the others in my group didn't want to".

One of the goals of these workshops have been to demonstrate to the participants that they could consider widening repertoire of methods they could draw on to study the complexities between humans, objects, and systems. The card choosing activity enacted the design technique of brainstorming to generate concepts and ideas. To make tangible and demonstrate their design fiction prototypes, the participants were asked to have resorted to methods such as sketching, scenario-building, model-making and persona enactment. They have made use of standard craft and stationary materials such as pen and paper, and image clippings. However, during a couple of discussions after the workshops, participants asked questions like: "why are we doing this?" or, "what has this got to do with sociology?" We welcomed such questions, as they provided an opening for highlighting academic debates about futures and about creative methods in social research. These questions also gave us an opportunity to demonstrate that social research might include the use of "live methods" (Back and Puwar, 2012), inventive methods' (Lury and Wakeford, 2012) or speculative research (Wilkie et al., 2017) approaches. Using examples of topics, the participants were planning to do research on, we would try to imagine what questions and answers we might generate by using playful, hands-on, collaborative and embodied ways of engaging with research subjects and students during fieldwork and in the classroom.

6 Conclusion

This article contributes to discussions of creative methods, pedagogies, technological imaginaries and futures research, by highlighting the value of design fiction in social research. For the students and researchers, activists, artists who have been involved in these workshops, discussing moral, ethical and social questions surrounding technologies is hardly something new. however, the major question has been to explore what we can learn from inviting them to imagine themselves as makers of technology futures, making these futures tangible, and getting a "feel" for the future (Coleman and Lyon, 2023: 424) Imagining futures in a world of uncertainty regarding the consequences of rapid technological change is a challenging task. From our limited experience of running these workshops, we observe that participating in design fiction offers a temporary space for creative imagination. Elements of playfulness, humor, irony, and satire

serve as useful devices for participants to embody roles and weave narratives about the future scenarios that they are able to imagine.

When participants present the fictional worlds that they create in the workshops what becomes evident is the often-unacknowledged biases, assumptions and affective forces that populate their imagination. These moments enable them to get a glimpse of how "designing futures" is entangled with epistemological and ontological assumptions, and with political and affective relations (Macgilchrist et al., 2023). The process of crafting design fiction scenarios can enable self-reflexivity about the latent power asymmetries, assumptions, worldviews, values, and biases embedded in the technology futures that they create. The results of the workshop highlight a methodological challenge for research that seeks to challenge normative imaginaries and enable the creation of transformative future visions. The challenge is how to surpass the difficulty people face in imagining futures in ways that do not reproduce dominant societal narratives when asked to imagine possible futures (Markham, 2021; Pink, 2025).

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

Ethical review and approval was not required for the study on human participants in accordance with the local legislation and institutional requirements. Written informed consent from the patients/participants or patients/participants' legal guardian/next of kin was not required to participate in this study in accordance with the national legislation and the institutional requirements. Oral consent was given by all participants and the workshops were attended on a voluntary basis.

Author contributions

HM: Conceptualization, Investigation, Methodology, Funding acquisition, Writing – original draft. CB: Formal analysis, Writing – original draft, Investigation, Conceptualization, Methodology, Funding acquisition. BK: Methodology, Conceptualization, Investigation, Resources, Writing – original draft. LL: Methodology, Conceptualization, Investigation, Resources, Writing – original draft.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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The author(s) declared that generative AI was not used in the creation of this manuscript.

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